Numerical verification of the single-round security chain and the finite-size bound for random-slot hyperentangled direct communication

Companion to the manuscript v4_final

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Abstract

The manuscript certifies the smooth min-entropy of the raw message symbols of Protocol 2^{*} and Protocol 2^{**} against sequential attacks with quantum memory. The proof chains a local Bell-basis decomposition, source replacement (Prop. III.2), postselection-stable relabeling (Lemma III.3), an entropic uncertainty relation with quantum side information ((23)–(27), Prop. V.1), a calibrated receiver model (Sec. V), an affine min-tradeoff function (Sec. VI) and the generalised entropy accumulation theorem. This companion states four claims that the chain implies and that can be tested numerically, constructs simulation scenarios for each, and reports the outcome. The claims are: (C1) the random-slot schedule removes an undetected adaptive attack that the original schedule admits; (C2) every equality and inequality of the single-round chain holds against single-round attacks with full quantum side information; (C3) the hypotheses that separate a valid test from an invalid one are each necessary; (C4) the single-round bound $r(e)$ is attained and therefore cannot be improved as a function of the pooled mismatch. A fifth part (C5) checks the finite-size expressions. Random sampling of attacks is complemented by an adversarial search that minimises the gap $H(D|E',\mathrm{det}) - r(e)$ over isometries, and the completeness statement is evaluated exactly rather than only by Monte Carlo. All 38 numerical checks pass; three claims (C1, C3 for the unmixed bases, C4) are also proved analytically. Flowcharts F1–F5 document the simulation logic. Equations of this companion are numbered (S1), (S2), ...; plain numbers refer to the manuscript. Security against sequential attacks with memory is not simulated and continues to rest on the GEAT argument of the manuscript.

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1 Objectives

A security proof of this type cannot be confirmed by simulation, because its statement quantifies over all sequential attacks. Simulation can nevertheless answer three questions a referee will ask. First, does each finite-dimensional step of the proof hold when evaluated exactly? Second, are the hypotheses of the proof doing real work, that is, does removing any of them admit an explicit attack? Third, how far is the proved bound from what an adversary can actually achieve? The companion organises the simulations around five claims, each tied to specific statements of the manuscript.

C1 *Schedule.* Under the original rule of [19] the control statistics do not bound Eve's knowledge for an adaptive adversary; under Protocol 2* the same adversary is detected and the certificate is respected (Sec. II-B–C).

C2 *Single-round chain.* For every single-round attack and every fixed history, Prop. III.2 and Lemma III.3 hold with equality and (25), (27), (30), (41), (44) hold as inequalities, with Eve holding the complete purification and controlling the click probability.

C3 *Necessity of hypotheses.* Replacing $\mathcal{C}_1, \mathcal{C}_2$ by the unmixed bases, announcing J_A before the slot leaves the channel, ignoring the source-overlap correction, or using a basis-dependent detection efficiency without thinning each admits an explicit violation.

C4 *Tightness.* There is an admissible attack with $H(D|E', \mathrm{det}) = r(e)$ for every $e \in [0, 3/4]$.

C5 *Finite-size consistency.* Tables I–II, Prop. VII.1 and Prop. VIII.1 follow from (57)–(61); two improvements of the finite-size analysis are quantified.

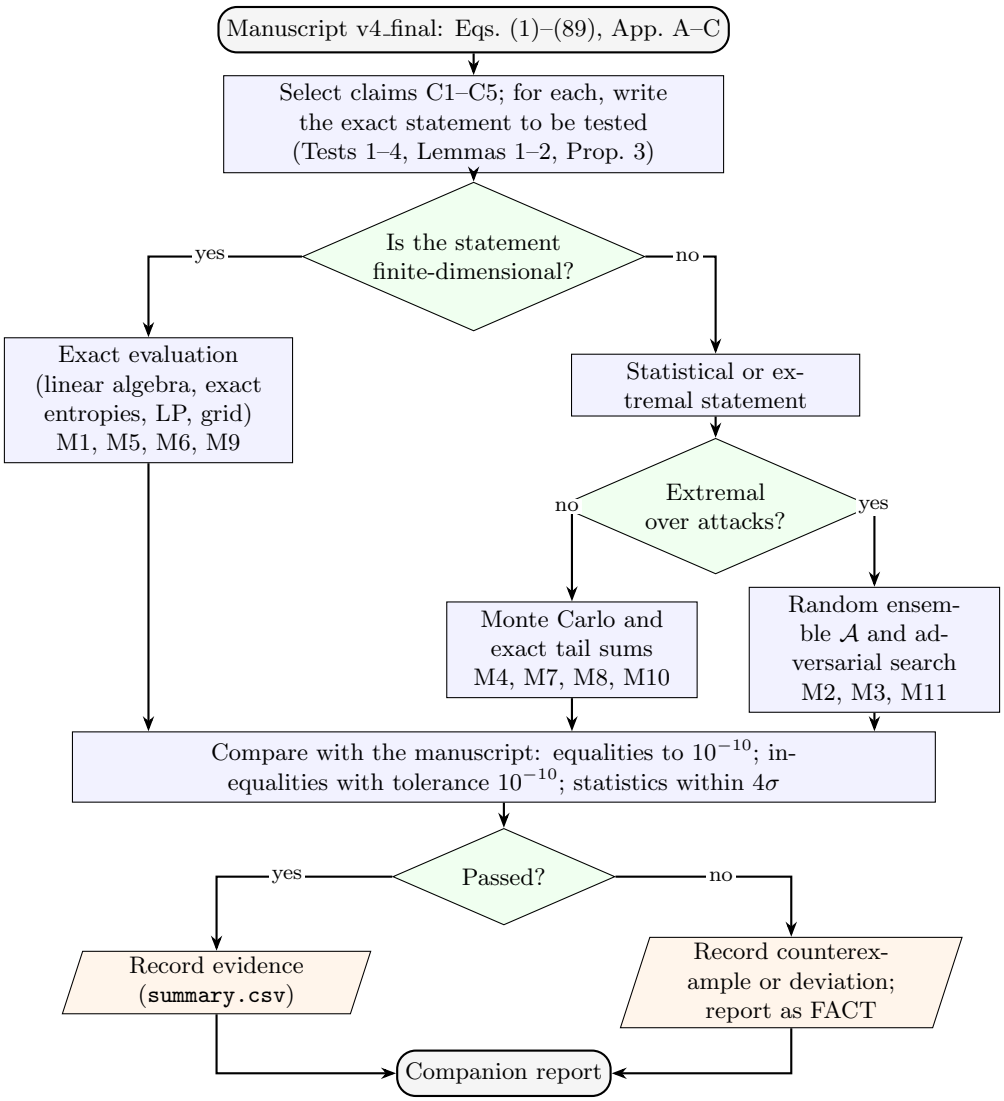


Figure 1: F1. Verification pipeline. Each claim is reduced to an exact statement and routed to exact evaluation, statistical simulation, or an extremal search over attacks.

Figure 1 shows how each claim is processed. A claim is first rewritten as an exact statement about computable quantities. Finite-dimensional statements are evaluated exactly. Statistical statements are evaluated by Monte Carlo and, where possible, by exact tail sums. Statements that quantify over attacks are evaluated in two complementary ways: over a random ensemble, which shows typical behaviour, and by an adversarial search, which looks for the worst case. A failed comparison is recorded as a counterexample rather than discarded.

2 Mathematical setting of the simulations

2.1 Spaces, bases and labels

Each photon carries polarization P and spatial mode S; its state space is $\mathbb{C}^4 = P \otimes S$ with ordered basis $(|H0\rangle, |H1\rangle, |V0\rangle, |V1\rangle)$. The local Bell detector basis is $\mathcal{D} = \{|d_{uv}\rangle\}_{u,v\in\mathbb{Z}_2}$ of (15)–(16), the mixed control bases are $\mathcal{C}_1 = Z_P \otimes X_S$ and $\mathcal{C}_2 = X_P \otimes Z_S$ of (10), and the unmixed bases are B_{ZZ}, B_{XX} of (6), (9). The message resource is

$$|\Phi\rangle_{AQ} = |\Phi^+\rangle_{P_A P_Q} |\Phi^+\rangle_{S_A S_Q} = \tfrac{1}{2} \sum_{k=0}^{3} |k\rangle_A |k\rangle_Q, \tag{S1}$$

where A is Alice's kept photon and Q the transmitted photon. Alice encodes the raw symbol $S = (a, w)$ by $Z^a X^w$ on P_A as in (2); J_A and J_B are the local detector labels and $D = J_A \oplus S$ is the virtual label (20). All bases used are real, so a virtual measurement in a basis and the preparation of the conjugate basis coincide.

2.2 Single-round attack and detection

Definition 1 (Admissible single-round attack)**.** An attack is an isometry $V : Q \to B \otimes E'$ with $B = \mathbb{C}^4 \oplus \mathbb{C}|\mathrm{vac}\rangle$ and E' finite-dimensional. Bob's trusted detection instrument (28) is $\mathcal{I} = \{\mathcal{I}_{\mathrm{det}}, \mathcal{I}_{\mathrm{loss}}\}$ with $\mathcal{I}_{\mathrm{det}}(\cdot) = \Pi \cdot \Pi$, Π the projector onto $\mathbb{C}^4 \subset B$. It acts before the analyzer and is basis independent; Eve controls Pr[det] only through the state she sends. Optionally a trusted local unitary W_B acts on B after detection (receiver misalignment, Prop. V.3).

Because E' is the Stinespring environment of V, Eve holds the full purification of the post-channel state. Conditioning on an arbitrary previous history only enlarges E', so a test over admissible attacks is a test of the single-round statement in the conditional form used by the GEAT step.

Virtual description. The postselected virtual state is

$$|\Psi\rangle_{ABE'} = \frac{(I_A \otimes \Pi V)|\Phi\rangle_{AQ}}{\sqrt{\Pr[\mathrm{det}]}}, \qquad \Pr[\mathrm{det}] = \big\|(I_A \otimes \Pi V)|\Phi\rangle\big\|^2. \tag{S2}$$

Alice's virtual message measurement is $\mathcal{D}$ on A with outcome D; her virtual test measurement is $\mathcal{C}_j$ on A with outcome U_j; Bob measures $W_B^\dagger \mathcal{C}_j W_B$ on B with outcome $\hat{U}_j$. The simulated quantities are

$$H(D|E', \mathrm{det}) = H\Big(\textstyle\sum_d |d\rangle\langle d| \otimes \mathrm{Tr}_{AB}\left[(|d\rangle\langle d|_A \otimes I)\Psi\right]\Big) - H(\rho_{E'}), \tag{S3}$$

$$e_j = \Pr[U_j \neq \hat{U}_j \mid \mathrm{det}], \qquad e = \tfrac{1}{2}(e_1 + e_2), \tag{S4}$$

that is, the pooled detected mismatch (26) with $\lambda = 1/2$. Entropies are von Neumann entropies computed from the eigenvalues of the cq blocks.

Actual description. A message round prepares $(Z^a X^w \otimes I)|\Phi\rangle$ for uniform (a, w), sends Q through V, applies $\mathcal{I}$, and Alice measures A in $\mathcal{D}$ to obtain J_A. A control round prepares a uniformly chosen $|c\rangle \in \mathcal{C}_j$ on Q, and Bob measures $\mathcal{C}_j$. The simulated quantities are $H(S|J_A E', \mathrm{det})$, $H(S|E', \mathrm{det})$, the actual mismatches e_j^{act} and the detection probabilities of message and control rounds.

2.3 Attack ensemble

The ensemble $\mathcal{A}$ consists of four randomised families and two explicit constructions. Parameters are drawn independently for each sample.

(a) *Coherent Haar mixture:* $V = \sqrt{1-f}\,(\iota \otimes |0\rangle_M) + \sqrt{f}\,(W \otimes |1\rangle_M)$, ι the embedding $Q \hookrightarrow \mathbb{C}^4 \subset B$, $W : Q \to B \otimes E_2$ Haar-random, $\dim E_2 \in \{1, \ldots, 4\}$, $f = u^2$, $u \sim U[0, 1]$. Loss enters through the vacuum component of W.

(b) *Coherent interaction with state-dependent loss:* $U = e^{-i\theta G}$ with G a GUE matrix on $Q \otimes E_0$, $\dim E_0 = 4$, $\theta \sim U[0, 3]$, followed by replacement of the photon by $|\mathrm{vac}\rangle$ with amplitude $\sin\phi$ conditioned on two of the four E_0 states, $\phi \sim U[0, 1.2]$, and exact re-isometrisation (QR).

(c) *Partial Bell measurement with outcome-dependent blocking:* Kraus operators $K_0 = \sqrt{1-f}\,I$, $K_d = \sqrt{f(1-\beta_d)}\,|d\rangle\langle d|$, $K'_d = \sqrt{f\beta_d}\,|\mathrm{vac}\rangle\langle d|$, $f, \beta_d \sim U[0, 1]$. This family targets loss postselection directly: Eve blocks the photon depending on what she learned.

(d) *Receiver misalignment:* with probability 0.3, one of (a)–(c) followed by $W_B = e^{-i\varepsilon G_B}$, $\varepsilon \sim U[0, 0.6]$.

The explicit constructions are the Bell-dephasing channel $\mathcal{V}_{\mathrm{deph}}(e)$ of Proposition 3 and the parity measurement $\mathcal{V}_{\mathrm{par}}$ of Lemma 2. In total 3000 random attacks are evaluated, 2054 of them without W_B and therefore also in the actual description.

2.4 Evaluation algorithm and the role of sample sizes

Figure 2 gives the algorithm of the single-round evaluation. For each sampled isometry the virtual quantities (S3)–(S4) and the actual quantities are computed from the same V, so every comparison is between two exact numbers and contains no sampling error. The sample size $N = 3000$ therefore controls only how much of the attack space is explored, not the accuracy of any reported value. A larger N cannot turn the random ensemble into a proof for all attacks; for that reason the worst case is addressed separately by the explicit construction of Proposition 3 and by the adversarial search of Sec. 6.1.

Monte Carlo sampling in the statistical sense appears only in the slot-level schedule model ($L = 2 \times 10^5$ slots, five seeds), the receiver estimators (4×10^5 to 2×10^6 rounds) and the completeness check (4000 runs per block length). In each case the target value has a closed form, so the sample size determines only the resolution. If no event is observed in N independent runs, the 95% upper confidence limit on its probability is approximately $3/N$, i.e. 7.5×10^{-4} for $N = 4000$. Probabilities of order 10^{-10} cannot be resolved by sampling at any practical N, which is why the abort probability is also computed exactly in Sec. 7.3.

2.5 Slot-level model for the schedule

For C1 each slot has type $T_i \in \{\mathrm{msg}, \mathrm{ctrl}\}$. The original rule of [19] is the Markov chain $\Pr[T_i = \mathrm{ctrl} \mid T_{i-1} = \mathrm{msg}] = p$, $\Pr[T_i = \mathrm{ctrl} \mid T_{i-1} = \mathrm{ctrl}] = 0$; Protocol 2* uses i.i.d. types with $\Pr[T_i = \mathrm{ctrl}] = p$. Control slots draw one of the sixteen states (6)–(9); only $\mathcal{C}_1, \mathcal{C}_2$ enter the test. The type of slot $i-1$ is public once slot i is sent. Eve's knowledge is counted classically: κ is the fraction of message slots in which her record determines S_i once J_A is public. For an attack without quantum memory, $\frac{1}{n}\sum_i H(S_i|Y_i, P^L) \leq 2(1-\kappa)$ with Y_i her record, so any valid first-order certificate must satisfy $r(e) \leq 2(1-\kappa)$. The simulation uses $L = 2 \times 10^5$ slots and five seeds.

3 Claim C1: the scheduling obstruction and its repair

Objective. Sec. II-B states in words that the original rule "can reveal that the next transmission is a message transmission" and that this "obstructs a direct random-sampling interpretation". The objective is to turn this into a quantitative statement for one concrete adaptive strategy and to show that Protocol 2* removes it.

Scenario. Strategy $\mathsf{S}_{\mathrm{adapt}}$: whenever slot $i-1$ is disclosed as a control slot, Eve measures slot i in $\mathcal{D}$ and resends the outcome; otherwise she does nothing. On a message slot this is non-disturbing, because the transmitted photon is a $\mathcal{D}$ eigenstate after Alice's measurement. A second scenario announces J_A before slot i leaves the channel.

Lemma 1 (Exact statistics of $\mathsf{S}_{\mathrm{adapt}}$)**.** *(i) Under the original rule, $e = 0$ and $\kappa = p$. (ii) Under Protocol 2*, $e = 3p/4$ and $\kappa = p$. (iii) If the slot type is public before the slot leaves the channel, Eve attacks exactly the message slots and $e = 0$, $\kappa = 1$.*

Proof. (i) No control slot follows a control slot, so Eve never touches a control slot and $e = 0$. The stationary control fraction is $\pi = p/(1+p)$, hence $\kappa = \Pr[T_{i-1} = \mathrm{ctrl} \mid T_i = \mathrm{msg}] = \pi/(1-\pi) = p$. (ii) With i.i.d. types, $\kappa = \Pr[T_{i-1} = \mathrm{ctrl}] = p$, and a mixed-basis control slot is attacked with

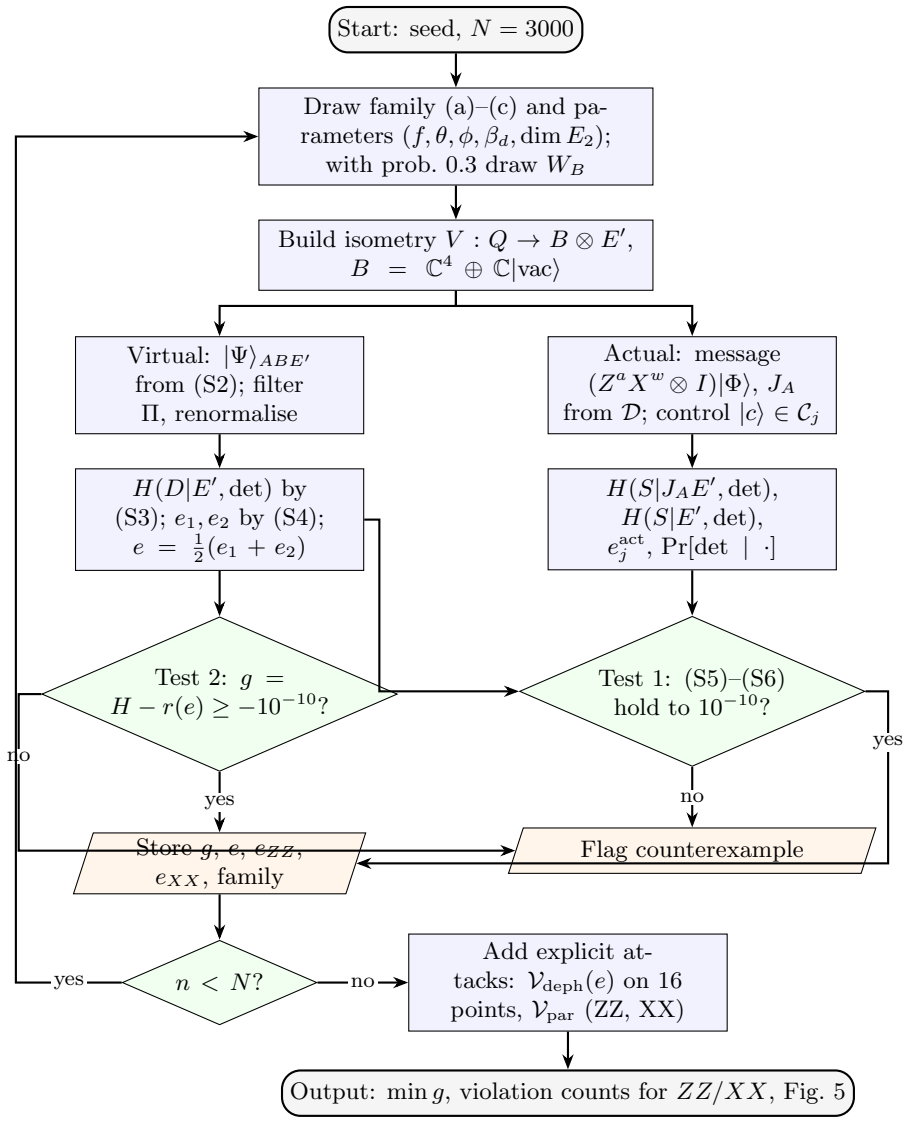


Figure 2: F2. Single-round evaluation (modules M2, M3). Both descriptions are evaluated for the same isometry; the equalities of Test 1 and the inequality of Test 2 are checked per sample.

probability p. A $\mathcal{D}$ measurement followed by resending on $|c\rangle \in \mathcal{C}_j$ returns the correct label with probability $\sum_d |\langle c|d\rangle|^4 = 4 \cdot (1/4)^2 = 1/4$ by (19), so the mismatch on attacked slots is $3/4$ and $e = 3p/4$. (iii) is immediate. □

Simulation method. Monte Carlo of the slot sequence with exact Born-rule sampling for every slot, for $p \in \{0.05, 0.1, \ldots, 0.5\}$; the algorithm is shown in Fig. 3.

Result. The original rule gives $e = 0$ and $\kappa = p$ to sampling accuracy, so the certified value $r(0) = 2$ exceeds the ceiling $2(1 - p)$ for every p. Protocol 2^* gives $e = 3p/4$ (largest deviation 0.0021) and $r(3p/4) < 2(1 - p)$ throughout, for example $r = 1.161$ against 1.600 at $p = 0.2$. Early announcement gives $e = 0$, $\kappa = 1$ (Fig. 4).

What is established. The obstruction of Sec. II-B is a concrete attack and not only a gap in the original argument; the random-slot repair is sufficient to expose this attack, and the timing of the J_A announcement is a separate necessary condition. The result concerns one strategy and the first-order rate; it is not a security statement for Protocol 2^*.

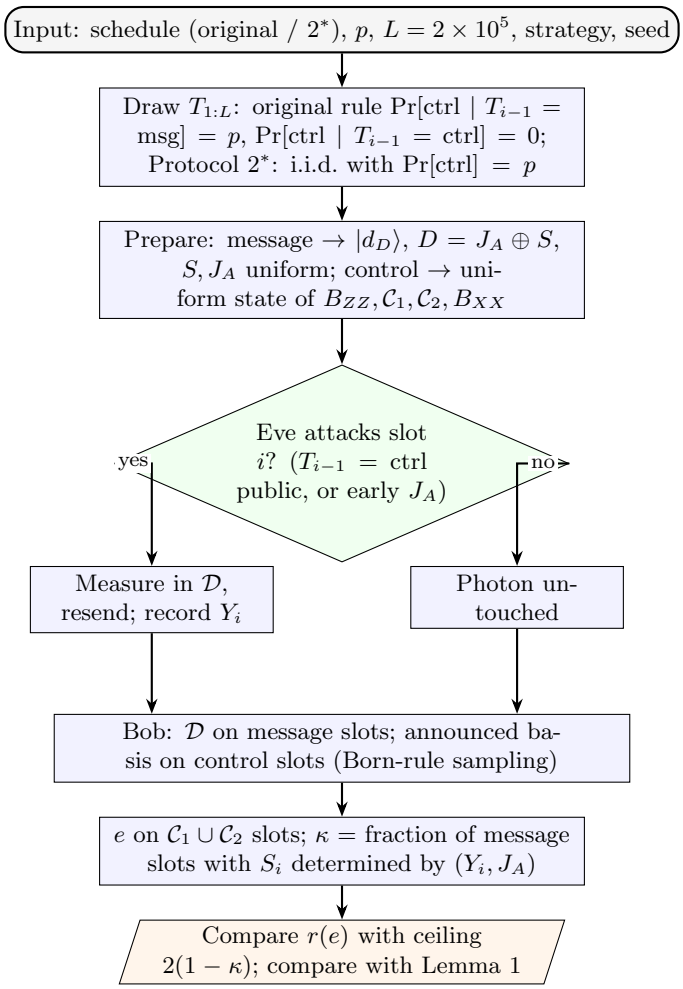


Figure 3: F4. Slot-level simulation of the scheduling obstruction (module M8). All slots are processed in vectorised form; the loop over i is shown for clarity.

4 Claim C2: the single-round chain

4.1 Optical core

Objective. Verify the finite-dimensional identities on which source replacement rests: (15)–(17) and App. A, Lemma III.1 (18), the detector relation (3), the virtual label (20), the overlap relation (19), and the statement of Sec. II-C and Remark II.1 that every ensemble averages to $I_4/4$.

Method. Exact linear algebra in double precision.

Result. The column matrix of $\mathcal{D}$ is unitary to 2.2×10^{-16} and its adjoint is (17). For all (a, w), (18) holds to 1.1×10^{-16}, every nonzero joint outcome satisfies $a = u \oplus u'$, $w = v \oplus v'$, J_A is uniform, and the conditional state of Q given J_A is $|d_{J_A \oplus S}\rangle$. All overlaps $|\langle d|c\rangle|^2$ with $c \in \mathcal{C}_1 \cup \mathcal{C}_2$ equal $1/4$, those with B_{ZZ}, B_{XX} take values $\{0, 1/2\}$, and $\mathcal{C}_1$, $\mathcal{C}_2$ are mutually unbiased. The reduced state of Q in every message round and the averages of the sixteen-state and eight-state control ensembles equal $I_4/4$ exactly.

4.2 Source replacement and relabeling

Test 1 (Prop. III.2 and Lemma III.3)**.** For every $V \in \mathcal{A}$ without W_B:

$$e_j^{\mathrm{act}} = e_j, \quad \Pr^{\mathrm{act}}[\mathrm{det} \mid \mathrm{msg}] = \Pr^{\mathrm{act}}[\mathrm{det} \mid \mathcal{C}_j] = \Pr[\mathrm{det}], \quad H(S|J_A E', \mathrm{det})^{\mathrm{act}} = H(D|E', \mathrm{det}), \tag{S5}$$

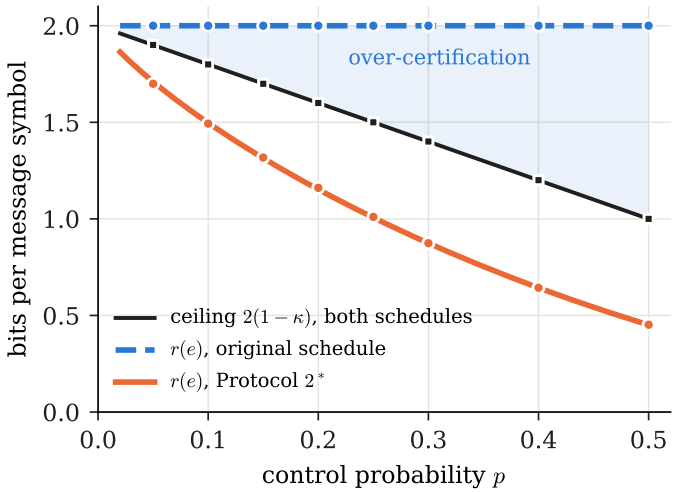


Figure 4: C1. Certified single-round rate $r(e)$ and the ceiling $2(1-\kappa)$ under $\mathsf{S}_{\mathrm{adapt}}$. Lines: Lemma 1; markers: Monte Carlo.

and, without conditioning on J_A,

$$H(S|E',\mathrm{det})^{\mathrm{act}} = 2. \tag{S6}$$

Scenario. The 2054 attacks of families (a)–(c), including attacks whose blocking probability depends on Eve's Bell outcome.

Method. Each quantity is computed exactly in both descriptions; no sampling is involved.

Result. The largest deviations in (S5) are 6.7×10^{-16} for mismatches and detection probabilities and 2.9×10^{-15} for the entropies. Equation (S6) holds to 1.8×10^{-15}.

What is established. The actual protocol and its virtual description produce identical classical–quantum statistics even when Eve exploits loss postselection, which is the content of Prop. III.2 under the common-filter hypothesis. Lemma III.3 transfers the entropy of D to S exactly. Equation (S6) shows that before J_A is public Eve's state is independent of the symbol for every attack: J_A acts as the key of a one-time pad with ciphertext D, so the security statement must condition on J_A and the moment of its announcement belongs to the protocol definition.

4.3 Uncertainty bound with loss and receiver imperfections

Test 2 ((25), (27), Prop. V.1, Prop. V.3)**.** For every $V\in\mathcal{A}$, including attacks with W_B,

$$H(D|E',\mathrm{det}) \;\geq\; r\big(\tfrac{1}{2}(e_1+e_2)\big), \qquad r(e)=2-h_2(e)-e\log_2 3 \;\; (0\leq e\leq 3/4), \;\; r(e)=0 \;\; (e>3/4). \tag{S7}$$

Scenario. All 3000 random attacks, of which 946 include a receiver unitary W_B.

Method. Exact evaluation of (S3)–(S4); the test statistic is the gap $g = H(D|E',\mathrm{det}) - r(e)$.

Result. $\min g = 2.4\times10^{-8}$ over all attacks and 1.5×10^{-5} over the attacks with W_B (Fig. 5a).

What is established. The single-round bound holds against full quantum side information, state-dependent loss and trusted local receiver imperfections. Because the minimum gap approaches zero, the test is not trivially satisfied; Section 6 shows that it is attained.

4.4 Source overlap

Test 3 (Prop. V.4, Remark V.5)**.** Let Alice's virtual bases be $\tilde{\mathcal{D}} = W_D\mathcal{D}$ and $\tilde{\mathcal{C}}_j = W_C\mathcal{C}_j$ with $W = e^{-i\varepsilon G}$, and $c_{\mathrm{src}} = \max_{j,n,z} |\langle \tilde{d}_n|\tilde{c}_z^{(j)}\rangle|^2$, $q_{\mathrm{src}} = \min\{2, -\log_2 c_{\mathrm{src}}\}$ as in (42). Then

$$H(\tilde{D}|E') \geq r_{q_{\mathrm{src}}}(e), \qquad c_{\mathrm{src}} \leq \min\{1, (\tfrac{1}{2}+\delta_D+\delta_C+\delta_D\delta_C)^2\}. \tag{S8}$$

Scenario. 1500 draws with $\varepsilon_D, \varepsilon_C \sim U[0, 0.5]$, giving $c_{\mathrm{src}} \in [0.26, 0.79]$; attacks from (a)–(c) and, as a fourth family, a partial measurement in the perturbed basis $\tilde{\mathcal{D}}$. Bob keeps the ideal analyzer. Rank-one bases always average to $I_4/4$, so the common-state hypothesis of Sec. V-F holds.

Result. The first inequality in (S8) holds with minimum gap 8.8×10^{-7}; the second with minimum slack 0.009.

4.5 Calibrated receiver

Objective and method. Check the three receiver reductions by exact algebra and Monte Carlo.

(i) *Thinning, Prop. V.2.* With efficiencies $\eta_z^{(y)} \sim U[0.55, 0.95]$ and $\eta_0 = \min \eta_z^{(y)} = 0.568$, the retained POVM $\theta_z^{(y)} M_z^{(y)}$ equals $\eta_0 \Pi_z^{(y)}$ to 5.6×10^{-17}. For a fixed input state the click probabilities of the $\mathcal{D}$, $\mathcal{C}_1$, $\mathcal{C}_2$ analyzers are 0.775, 0.618, 0.605 before thinning and 0.568, 0.569, 0.569 after (4×10^5 samples each); the retained outcome distribution matches the ideal projective one within total variation 1.4×10^{-3}.

(ii) *Dark events, (34)–(36).* For $p_{\mathrm{dc}} \in \{0, 10^{-3}, 10^{-2}, 5\times10^{-2}\}$ and 2×10^6 rounds, the estimators $\eta_{\mathrm{sig}} = 1-p_{\mathrm{dc}}-q'_{\mathrm{loss}}$ and $e_{\mathrm{sig}} = (q'_{\mathrm{err}}-3p_{\mathrm{dc}}/4)/\eta_{\mathrm{sig}}$ are unbiased within four standard deviations; at $p_{\mathrm{dc}} = 0.05$, $e_{\mathrm{sig}} = 0.050$ is estimated as 0.0501 ± 0.0004, while the uncorrected detected mismatch is 0.114.

(iii) *Cross-talk, (38)–(40).* The relation $e_{\mathrm{obs}} = \chi+(1-4\chi/3)e$ holds exactly and within 1.1×10^{-3} on sampled labels. The linear program (39) with exact equality returns the inverted intrinsic mismatch to 3×10^{-16}; with a confidence box of half-width 10^{-3} or 10^{-2} it returns 0.031 or 0.040 for $e = 0.03$, $\chi = 0$, so a finite-data region produces the upward margin required by the manuscript.

5 Claim C3: necessity of the hypotheses

Objective. Show that each hypothesis separating a valid secrecy test from an invalid one is necessary, by exhibiting an admissible attack that violates the corresponding bound once the hypothesis is removed.

5.1 Unmixed bases are not valid secrecy tests

Lemma 2 (Parity attack)**.** *Let Eve measure the observable $Z_P \otimes Z_S$ non-destructively, i.e. apply the instrument with Kraus operators $\Pi_\pm = \frac{1}{2}(I \pm Z_P \otimes Z_S)$ and keep the outcome. Then $e_{ZZ} = 0$, $H(D|E') = 1$, and $e_1 = e_2 = 1/2$. Hence $H(D|E') < r(e_{ZZ}) = 2$, while $H(D|E') \geq r(1/2) = 1 - \frac{1}{2}\log_2 3 \approx 0.208$ is respected.*

Proof. Π_+ projects onto $\mathrm{span}\{|H0\rangle, |V1\rangle\} = \mathrm{span}\{|d_{00}\rangle, |d_{10}\rangle\}$, so $Z_P \otimes Z_S |d_{uv}\rangle = (-1)^v |d_{uv}\rangle$ and the outcome equals the bit v of D; the bit u remains uniform and unknown, giving $H(D|E') = 1$. $Z_P \otimes Z_S$ is diagonal in B_{ZZ}, so the instrument leaves every B_{ZZ} state unchanged and $e_{ZZ} = 0$. On $\mathcal{C}_1 = Z_P \otimes X_S$ the observable flips the X_S eigenvalue, so the post-measurement state is an equal mixture of the sent state and its partner, and $e_1 = 1/2$; likewise $e_2 = 1/2$. □

The Hadamard-rotated attack ($X_P \otimes X_S$, revealing u) gives $e_{XX} = 0$ and $H(D|E') = 1$. Numerically, among the 3000 random attacks 943 violate $H \geq r(e_{ZZ})$ and 940 violate $H \geq r(e_{XX})$, with violations up to 0.35 bit (Fig. 5b). This confirms the statement after (10) and raises a point the manuscript leaves open: if B_{ZZ} and B_{XX} are retained as "auxiliary diagnostic checks", their role in the abort decision should be stated. FACT{abort rule for the ZZ/XX diagnostics in Sec. II-C}

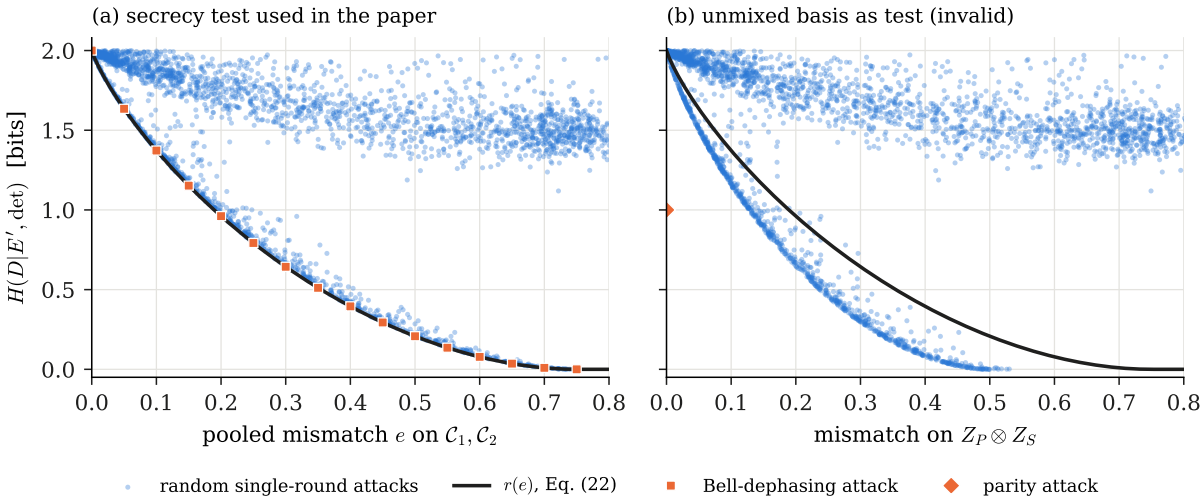


Figure 5: C2–C4. $H(D|E', \mathrm{det})$ for 3000 random single-round attacks. (a) Against the pooled $\mathcal{C}_1, \mathcal{C}_2$ mismatch: all points lie above $r(e)$ and the Bell-dephasing attack (squares) lies on it. (b) The same attacks against the $Z_P \otimes Z_S$ mismatch: many points and the parity attack (diamond) lie below $r(e)$.

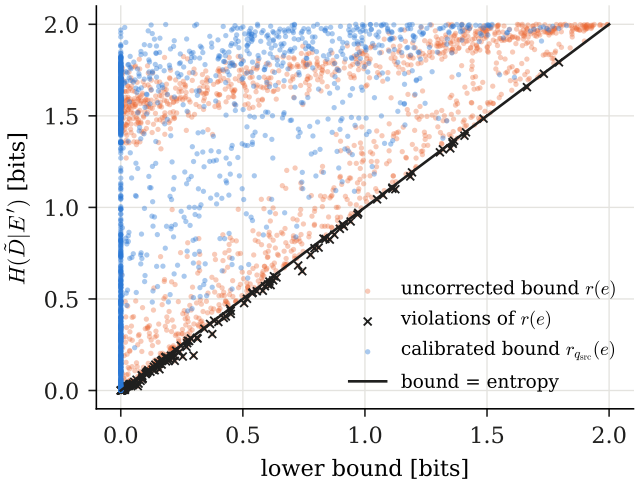


Figure 6: C3. Entropy versus lower bound for perturbed source bases. Violations (crosses) occur only for the uncorrected bound.

5.2 Announcement timing

Lemma 1(iii) and (S6) together show that the hypothesis "the channel and instrument do not act on J_A" of Lemma III.3 is necessary in its operational form: if J_A, or any public datum that reveals the slot type, is available before the slot leaves the channel, Eve attacks message slots only, $e = 0$ and $\kappa = 1$.

5.3 Source-overlap correction

In the fourth family of Test 3, where Eve measures partially in the perturbed source basis $\tilde{\mathcal{D}}$, 144 of 375 cases violate the uncorrected bound $H(\tilde{D}|E') \geq r(e)$ by up to 0.107 bit, while none violates $r_{q_{\mathrm{src}}}(e)$ (Fig. 6). The reduction of the complementarity constant from 2 to q_{src} in Prop. V.4 and Corollary VI.3 is therefore necessary.

5.4 Basis-dependent detection

Before thinning, the click probabilities of the three analyzers differ by 0.17 (Sec. 4.5(i)). A basis-dependent filter breaks the equality $\Pr[\text{det} \mid \text{msg}] = \Pr[\text{det} \mid \mathcal{C}_j]$ used in Prop. V.1, and thinning restores it to within 1.6×10^{-3}. The hypothesis is therefore not automatically satisfied by a calibrated receiver and has to be enforced.

6 Claim C4: tightness of the single-round bound

Proposition 3 (Bell-dephasing attack)**.** *For $m = (m_1, m_2) \in \mathbb{Z}_2^2$ let $Z_m = \sum_d (-1)^{m\cdot d} |d\rangle\langle d|$. Then, up to phases, $Z_{00} = I$, $Z_{01} = Z_P \otimes Z_S$, $Z_{10} = X_P \otimes X_S$, $Z_{11} = Y_P \otimes Y_S$. For $p = (1-e, e/3, e/3, e/3)$, $0 \le e \le 3/4$, the channel with Kraus operators $\sqrt{p_m}\, Z_m$ and Stinespring environment $E' \cong \mathbb{C}^4$ satisfies* $\Pr[\text{det}] = 1$, *$e_1 = e_2 = e$, and*

$$H(D|E') = 2 - H(p) = 2 - h_2(e) - e\log_2 3 = r(e). \tag{S9}$$

Proof. The identification of Z_m follows from $Z_{01}|d_{uv}\rangle = (-1)^v |d_{uv}\rangle$ and $Z_{10}|d_{uv}\rangle = (-1)^u |d_{uv}\rangle$ in (15)–(16). Each nontrivial Z_m flips the eigenvalue of at least one tensor factor of $\mathcal{C}_1 = Z_P \otimes X_S$ and of $\mathcal{C}_2 = X_P \otimes Z_S$, so it permutes each mixed basis without fixed points; hence $e_j = 1 - p_{00} = e$. The joint state is $\sum_m \sqrt{p_m}(I \otimes Z_m)|\Phi\rangle|m\rangle_{E'}$. Since $(\langle d|_A \otimes I)|\Phi\rangle = \frac{1}{2}|d\rangle_Q$, conditioning on $D = d$ gives $\frac{1}{2}|d\rangle_B|\varphi_d\rangle_{E'}$ with $|\varphi_d\rangle = \sum_m \sqrt{p_m}(-1)^{m\cdot d}|m\rangle$. Then $\rho_{E'} = \frac{1}{4}\sum_d |\varphi_d\rangle\langle\varphi_d|$ has entries $\sqrt{p_m p_{m'}}\,\frac{1}{4}\sum_d (-1)^{(m\oplus m')\cdot d} = p_m \delta_{mm'}$, and the conditional states are pure, so $H(D|E') = H(D) - H(\rho_{E'}) = 2 - H(p)$. □

Numerical confirmation. Evaluating $\mathcal{V}_{\text{deph}}(e)$ at sixteen values of $e \in [0, 3/4]$ gives $|H(D|E') - r(e)| \le 6.7 \times 10^{-16}$ (squares in Fig. 5a).

What is established. The channel is an admissible attack (a random two-qubit Pauli channel on Q with no loss). Therefore no valid single-round bound that depends on the test data only through the pooled mismatch e can exceed $r(e)$ at any $e \in [0, 3/4]$. Any improvement of the certified rate must use additional statistics or come from the finite-size terms; a semidefinite relaxation over the same pooled statistic cannot improve (27).

6.1 Adversarial search over isometries

Test 4 (Extremal form of (27))**.** For every $\dim E'$,

$$\inf_{V:\, Q\to B\otimes E'} \Big[H(D|E', \text{det}) - r\big(\tfrac{1}{2}(e_1 + e_2)\big) \Big] \;\ge\; 0, \tag{S10}$$

and by Proposition 3 the infimum equals 0 once $\dim E' \ge 4$.

Objective. Random sampling explores typical attacks; a referee may ask whether a carefully optimised attack does better. The objective is to search for the worst attack directly and to calibrate the search on a test that is known to be invalid.

Scenario. Every isometry is written as $V = X(X^\dagger X)^{-1/2}$ with an unconstrained $X \in \mathbb{C}^{5\dim E' \times 4}$; the polar map is smooth and onto the set of isometries. The objective is

$$J(X) = H(D|E', \text{det}) - r(e) + w\,(e - e^*)^2 + 100\,\big[\max\{0,\, 0.05 - \Pr(\text{det})\}\big]^2, \qquad w = 200, \tag{S11}$$

where the penalty steers the search to a target mismatch e^* and excludes the degenerate case of almost certain loss. Targets $e^* \in \{0.02, 0.05, 0.1, 0.2, 0.3, 0.45, 0.6\}$ are used with $\dim E' = 2$ (two starts each), $\dim E' = 4$ (three starts) and $\dim E' = 8$ (three targets). As a control, the same objective with e replaced by the $Z_P \otimes Z_S$ mismatch is minimised for $e^* \in \{0, 0.05, 0.1, 0.2\}$ and $\dim E' \in \{2, 4\}$.

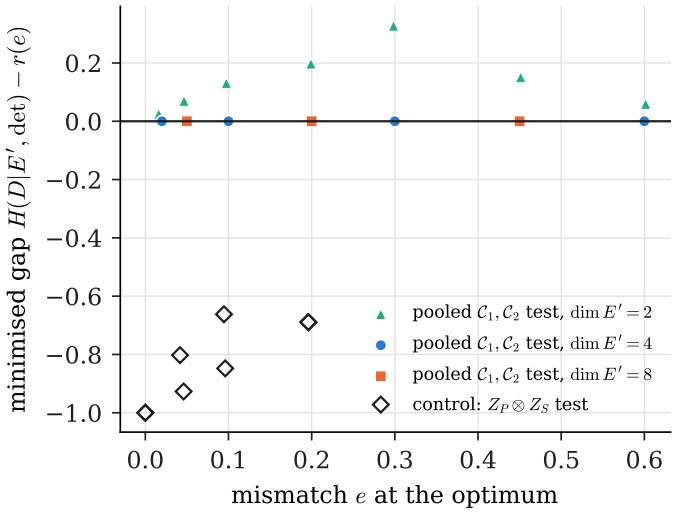


Figure 7: C4. Gap at the end of each adversarial search versus the reached mismatch. Filled markers: pooled $\mathcal{C}_1, \mathcal{C}_2$ test; open diamonds: control search against the invalid $Z_P \otimes Z_S$ test.

Method. L-BFGS-B on $(\mathrm{Re}\, X, \mathrm{Im}\, X)$ with finite-difference gradients, at most 600 iterations, starting from the identity channel with Gaussian perturbation (Fig. 8). Each start takes 3–20 s; the full search takes about five minutes.

Result. Over 38 optimised attacks against the pooled test the smallest gap is 1.8×10^{-8}, and no start produces a negative gap (Fig. 7). For $\dim E' = 4$ and 8 every start converges to the bound, with largest final gap 1.4×10^{-5}, and the optimal entropies reproduce $r(e)$ at all targets, e.g. $H = 1.37251$ at $e = 0.100$ against $r(0.1) = 1.37251$. For $\dim E' = 2$ the optimum stays strictly above the bound, with gaps between 0.025 and 0.33. The control search finds gaps between -1.000 and -0.66 against the $Z_P \otimes Z_S$ test, recovering the parity attack of Lemma 2 at $e^* = 0$ and exceeding the largest random violation (-0.35) by a factor of about three.

What is established. A worst-case search that demonstrably finds violations when they exist finds none for the test used in the manuscript, and it converges from independent random starts to the value $r(e)$ predicted by Proposition 3. The search also indicates that an adversary needs a single-round memory of dimension at least four to saturate (27) within the searched class; this is an observation from a finite search, not a proven threshold. NEXT STEP{state the $\dim E' \geq 4$ observation only if it is proved, e.g. from the Kraus rank of optimal attacks}

7 Claim C5: finite-size consistency

7.1 GEAT ingredients

Objective and method. Check on dense grids and by linear programming the properties of the affine trade-off that enter (57)–(61): the global tangent (47), the sign conditions $\beta_t - \Lambda_t \leq 0$ and $\beta_t \leq 3\Lambda_t/4$, the vertex range $\mu\Lambda_t$ of (48) and App. B, the variance bound (52), and the worst accepted value (55), (74).

Result. On 400 tangent points in $[0.002, 0.74]$ and 20 001 values of $e \in [0, 1]$, $\min_e[r(e) - (\beta_t - \Lambda_t e)] = -1.1 \times 10^{-16}$, $\max(\beta_t - \Lambda_t) = -0.019$ and $\max(\beta_t - 3\Lambda_t/4) = -3.8 \times 10^{-4}$; the r_q tangent of Corollary VI.3 is also global. For five parameter sets, $\mathrm{Max}(f_t) - \mathrm{Min}_\Sigma(f_t) = \mu\Lambda_t$, the largest variance over the test simplex is 0.90–0.98 of $\mu^2\Lambda_t^2/\gamma$, and the minimum of f_t over the acceptance polytope (53)–(54) or (72)–(73) equals $\mu(1 - p_{\mathrm{dc}})\eta_{\mathrm{min}} r(e_{\mathrm{tol}})$ to 2.2×10^{-16}.

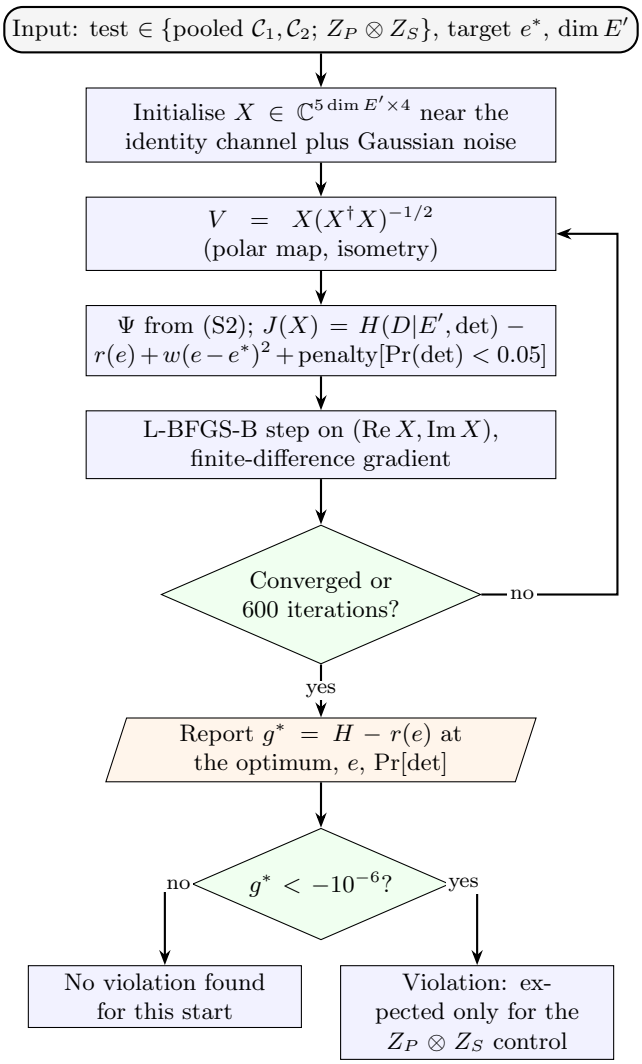


Figure 8: F3. Adversarial search (module M11). The same optimiser is run against the valid test and against the invalid $Z_P \otimes Z_S$ test, which calibrates its ability to find violations.

7.2 Tables I–II, symmetric bias and scaling

Method. Maximise $L^{-1} \times (58)$ over $p \in (0, 1)$ and $\alpha \in (1, 3/2)$ with $\varepsilon_s = \varepsilon_a = 10^{-10}$, using a logarithmic grid in p followed by Nelder–Mead refinement and a bounded search in $\log(\alpha - 1)$ (Fig. 11, left).

Result. All 18 entries of Table I and all 16 entries of Table II are reproduced to three decimals. For Protocol 2** at fixed $\mu \in \{0.5, 0.7, 0.85\}$, the maximiser on the curve $(1 - p_A)(1 - p_B) = \mu$ lies within the scan resolution of $p_A = p_B = 1 - \sqrt{\mu}$, and a free two-dimensional optimisation from three asymmetric starts converges to $p_A = p_B = 0.1985$ at $L = 10^6$, $e_{\mathrm{tol}} = 2\%$ (Prop. VII.1 on the full bound). For $L = 10^5$–10^{11} the fitted exponents of p_{opt}, $\alpha_{\mathrm{opt}} - 1$ and the deficit $r(e_{\mathrm{tol}}) - H^{\varepsilon}_{\min}/L$ are -0.335, -0.662, -0.337 for 2* (predicted $-1/3$, $-2/3$, $-1/3$) and -0.253, -0.761, -0.245 for 2** (predicted $-1/4$, $-3/4$, $-1/4$). The ratio of the full deficit to the leading term (87) lies in $[1.016, 1.052]$ for 2* and rises from 0.73 to 0.99 for 2** (Fig. 9).

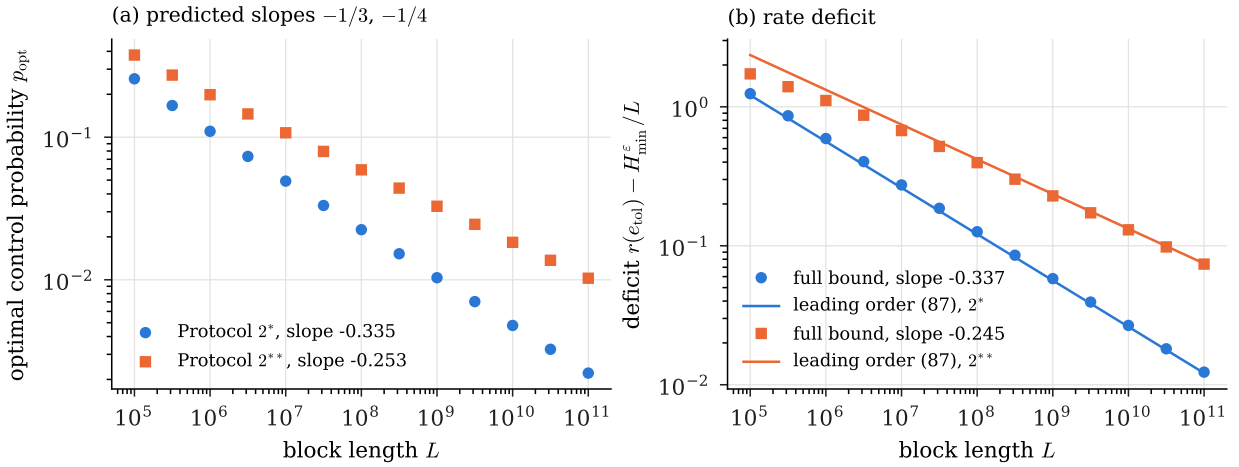


Figure 9: C5. (a) Optimal control probability and (b) deficit of the full bound (markers) against the leading-order expansion (87) (lines), $e_{\mathrm{tol}} = 2\%$.

7.3 Two quantified improvements

Free tangent point. Theorem VI.2 fixes $e_t = e_{\mathrm{tol}}$. For $p_{\mathrm{dc}} = 0$ the minimum of the lifted function over the acceptance polytope at a general tangent point is

$$\min_{\hat{q}\in\Omega} f_t(\hat{q}) = \mu\big[\beta_t - \max\{\Lambda_t e_{\mathrm{tol}},\ \beta_t(1-\eta_{\min}) + \Lambda_t e_{\mathrm{tol}}\eta_{\min}\}\big], \tag{S12}$$

which reduces to $\mu\eta_{\min} r(e_{\mathrm{tol}})$ at $e_t = e_{\mathrm{tol}}$. Using (S12) with Λ_t in V_G and X_G raises the bound at $L = 10^5$ from 0.564 to 0.781 (2*, $e_{\mathrm{tol}} = 1\%$, optimal $e_t = 0.134$) and from 0.077 to 0.284 (2**, $e_t = 0.205$); at $L = 10^6$, $e_{\mathrm{tol}} = 2\%$ the gains are 0.055 and 0.105 (Table 1). The gain is large when the variance term dominates, because a larger e_t lowers Λ_t. NEXT STEP{restate Theorem VI.2 for a general tangent point and confirm that every constant in (57), (61) uses Λ_t}

Table 1: $H^{\varepsilon}_{\min}/L$ with $e_t = e_{\mathrm{tol}}$ and with e_t optimised, ideal devices.

L	e_{tol}	Protocol	$e_t = e_{\mathrm{tol}}$	e_t free	e_t^{opt}
10^5	1%	2*	0.564	0.781	0.134
10^5	1%	2**	0.077	0.284	0.205
10^5	2%	2*	0.584	0.747	0.144
10^5	2%	2**	0.100	0.265	0.214
10^6	2%	2*	1.236	1.291	0.076
10^6	2%	2**	0.717	0.822	0.126
10^6	5%	2*	1.126	1.157	0.106
10^6	5%	2**	0.668	0.733	0.155

Completeness. For an honest i.i.d. channel the two accepted statistics are sums of bounded variables with variances $v_1 = p_\ell(1-p_\ell)$, $p_\ell = \gamma(1-p_{\mathrm{dc}})(1-\eta_{\mathrm{hon}})$, and $v_2 = \mathrm{Var}(\not{\mathbb{1}}_{\mathrm{err}} + e_{\mathrm{tol}}\not{\mathbb{1}}_{\mathrm{loss}})$, both of order γ, whereas (69) treats them as variables of range one. Bernstein's inequality with the same union bound gives

$$\varepsilon_{\mathrm{comp}} \le \sum_{k=1,2} \exp\Big[-\frac{Lt_k^2}{2v_k + 2t_k/3}\Big], \quad t_1 = \gamma(1-p_{\mathrm{dc}})(\eta_{\mathrm{hon}} - \eta_{\min}), \quad t_2 = \gamma(1-p_{\mathrm{dc}})\eta_{\mathrm{hon}}(e_{\mathrm{tol}} - e_{\mathrm{hon}}). \tag{S13}$$

The exact abort probability follows from the fact that the acceptance event (53)–(54), or (72)–(73), depends only on the counts N_ℓ of test-branch losses and N_e of recorded test errors, which

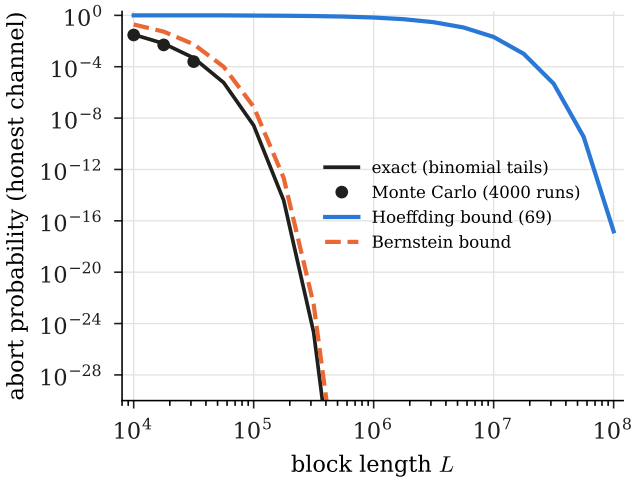


Figure 10: C5. Honest-channel abort probability for 2^*, $p = 0.11$: exact value (S14), Monte Carlo (points with at least one abort in 4000 runs), and the two bounds.

are multinomial with probabilities p_ℓ and $p_e = \gamma[(1 - p_{\mathrm{dc}})\eta_{\mathrm{hon}} e_{\mathrm{hon}} + 3p_{\mathrm{dc}}/4]$. With $A = \lfloor L\gamma(1 - p_{\mathrm{dc}})(1 - \eta_{\mathrm{min}})\rfloor$ and $b = \gamma[e_{\mathrm{tol}}(1 - p_{\mathrm{dc}}) + 3p_{\mathrm{dc}}/4]$,

$$\begin{aligned} \Pr[\mathrm{abort}] = \Pr[N_\ell > A] + \sum_{n=0}^{A} \Pr[N_\ell = n]\ \Pr\big[N_e > \lfloor Lb - e_{\mathrm{tol}} n\rfloor \ \big|\ N_\ell = n\big], \\ N_e \,|\, N_\ell = n \ \sim\ \mathrm{Bin}\Big(L - n, \tfrac{p_e}{1-p_\ell}\Big), \end{aligned} \tag{S14}$$

which is evaluated in log space and is exact down to the double-precision limit of about 10^{-300} (Fig. 11, right).

With $\eta_{\mathrm{hon}} = 0.8$, $e_{\mathrm{hon}} = 1\%$, $\eta_{\mathrm{min}} = 0.7$ and $e_{\mathrm{tol}} = 2\%$, the exact values agree with 4000 Monte Carlo runs per L within 1.5σ for 2^* ($p = 0.11$, with and without $p_{\mathrm{dc}} = 10^{-3}$) and 2^{**} ($p = 0.2$), and both (69) and (S13) lie above the exact value at every L. The exact computation turns a qualitative statement into a quantitative one (Fig. 10). For 2^*, the abort probability reaches 10^{-3} at $L = 2.7 \times 10^4$; the Bernstein bound reaches it at 4.2×10^4 (factor 1.5), the Hoeffding bound (69) at 1.8×10^7 (factor 650). For 2^{**} the corresponding block lengths are 7.6×10^4, 1.2×10^5 (factor 1.5) and 1.3×10^8 (factor 1780). At $L = 10^5$, where (69) still gives 0.96, the exact abort probability for 2^* is 2.7×10^{-9}. The manuscript allows a sharper bound after (69); with (S13) the completeness statement is within a factor 1.5 in block length of the exact value and becomes informative at the block lengths of Tables I–II.

7.4 Leftover hashing (classical special case)

For $n = 12$ uniform bits and four deterministic leakages $E = \phi(S)$ with exact $H_{\mathrm{min}}(S|E) = k \in \{6, 8, 8.30, 9\}$, the seed-averaged trace distance of $F(S)$ from uniform, computed exactly over 400 random binary $\ell \times n$ matrices per ℓ, never exceeds $\frac{1}{2}\, 2^{-(k-\ell)/2}$ in (65); the largest ratio over 48 pairs is 0.98. Only the classical form of the leftover-hash lemma is exercised; the quantum statement used in Theorem VI.5 is taken from [22].

8 Scope

The simulations establish that the single-round chain is exact where it claims equality and valid where it claims inequality against single-round attacks with full quantum side information, state-dependent loss and receiver imperfections (C2); that the hypotheses separating valid from invalid

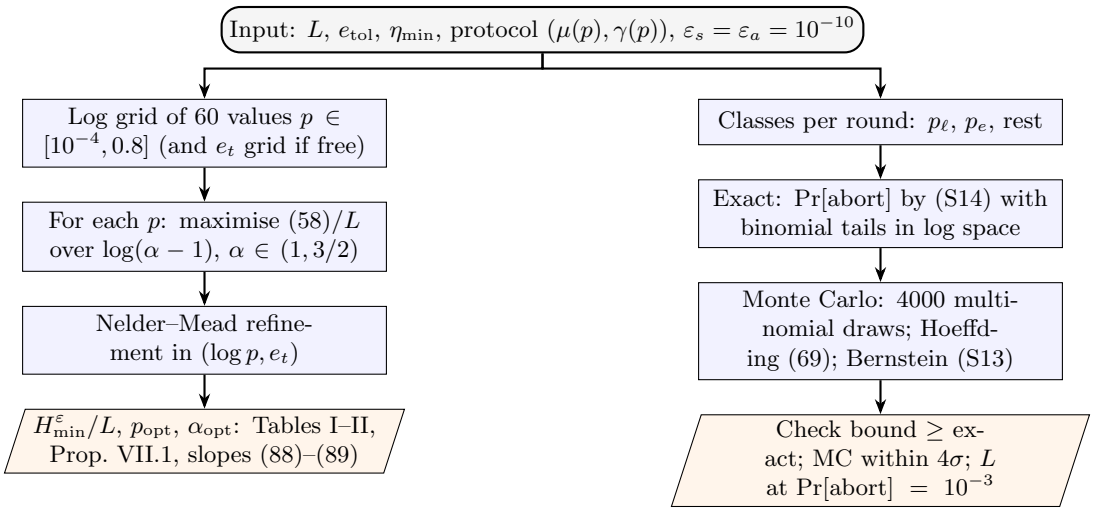


Figure 11: F5. Finite-size optimisation (module M6, left) and completeness evaluation (modules M7, M10, right).

tests are necessary (C3); that the bound $r(e)$ is attained, and that an adversarial search which finds violations of invalid tests finds none of (27) (C4); that the original schedule admits an undetected adaptive attack which Protocol 2^* exposes (C1); and that the finite-size results follow from the stated formulas (C5).

They do not establish security against sequential attacks with quantum memory. That statement depends on Lemma VI.1 and Appendix C, which are structural and were checked only by comparison with Theorem 4.3 and Lemma 4.7 of [25]. The random ensemble $\mathcal{A}$ samples but does not exhaust the single-round attacks, and the adversarial search is a local optimisation that cannot certify a global minimum; validity for all attacks rests on the analytical proof, and tightness on the explicit construction of Proposition 3. Multiphoton components, double clicks, finite calibration data and active detector control are outside the model, as they are outside the manuscript.

Use in the revision. Figure 5 and Lemma 2 support the choice of test bases in Sec. II-C; Fig. 4 and Lemma 1 support Sec. II-B; (S6) supports the treatment of J_A; Proposition 3 answers in advance the question whether the single-round bound is loose. NEXT STEP{decide which items enter the main text and which go to the supplementary material}

A Reproducibility

The code is a self-contained Python 3 package (`numpy`, `scipy`, `pandas`, `matplotlib`) with seeded generators. `python3 run_all.py` executes every check in about six minutes on one core, of which five are the adversarial search; `python3 run_all.py --quick` reuses a stored search result and runs in about one minute. Results go to `results/` and `figures/`. To show that conclusions do not depend on one random sequence, change the `seed` arguments of the modules; increasing the sample sizes improves resolution only (Sec. 2.4).

```
core.py           bases, entropies, r(e), r_q(e), tangent (46), worst accepted value, Delta_L (57)-(61)
m1_optics.py      Sec. 4.1   (15)-(20), App. A, (19), average states
m2_quantum.py     Tests 1-2, Lemma 2, Prop. 3   attack ensemble, actual vs virtual, H(D|E',det)
m3_source.py      Test 3, Sec. 5.3              perturbed virtual bases, (42)-(45)
m4_receiver.py    Sec. 4.5                      thinning, dark estimators, cross-talk and LP (39)
m5_tradeoff.py    Sec. 7.1                      tangent, vertex range, variance, acceptance LP
m6_finite.py      Sec. 7.2-7.3                  Tables I-II, Prop. VII.1, VIII.1, free tangent point
m7_complete.py    Sec. 7.3                      abort probability, (69), (75)-(76), Bernstein
m8_schedule.py    Lemma 1                       adaptive attack, early J_A, sifting rates (11)-(12)
m9_hash.py        Sec. 7.4                      exact classical leftover-hash check
m10_exact_abort.py  Eq. (S14)                   exact abort probability by binomial tails
```

```
m11_adversarial.py  Test 4, Fig. F3            adversarial search over isometries, ZZ control
run_all.py          Fig. F1                    executes all modules, writes the summary of App. B
```

The core single-round evaluation:

```
psi = einsum("beq,aq->abe", V, PHI16.reshape(4, 4))[:, :4, :]    # (I x Pi V)|Phi>, Eq. (2)
psi /= norm(psi)                                                  # condition on det
blocks = [(phi := einsum("a,abe->be", d.conj(), psi)).T @ phi.conj() for d in BELL]
H = cond_entropy_cq(blocks)                                       # H(D|E',det), Eq. (3)
P_j[k, l] = sum_e |<c_k|_A <c_l|_B psi|^2 ;  e_j = 1 - trace(P_j)  # Eq. (4)
assert H >= r(0.5 * (e_1 + e_2)) - 1e-10                          # Test 2
```

B List of numerical checks

Check	Result	Evidence
M1 App. A / (17) unitary and adjoint	pass	max dev 2.2e-16
M1 (19) mixed bases MUB with D; ZZ, XX not	pass	{'ZZ': [0.0, 0.5], 'ZX': [0.25], 'XZ': [0.25], 'XX': [0.0, 0.5]}
M1 Lemma III.1 (18) and relation (3)	pass	max dev 1.1e-16
M1 (20) D = J_A xor S	pass	all 16 (S, J_A)
M1 Sec. II-C / Rem. II.1 average states I/4	pass	message, 16-state, 8-state ensembles
M2 Prop. III.2 actual = virtual statistics	pass	2054 attacks, max dev 6.7e-16
M2 Lemma III.3 H(S|J_A E,det) = H(D|E,det)	pass	max dev 2.9e-15
M2 H(S|E,det) = 2 without J_A (no-signalling)	pass	max dev 1.8e-15
M2 (25)/(27)/Prop. V.1 H(D|E,det) $\geq$ r(e_pool)	pass	3000 attacks, min gap 2.42e-08
M2 Prop. V.3 receiver misalignment	pass	946 attacks, min gap 1.52e-05
M2 tightness: Bell dephasing attains r(e)	pass	max |gap| 6.7e-16 on 16 points
M2 counterexample: ZZ/XX test invalid	pass	parity attack e_ZZ=0, H=1 < r(0)=2; random attacks min gap_ZZ -0.349
M3 Prop. V.4 H(D$\sim$|E) $\geq$ r_q(e)	pass	1500 cases, min gap 8.83e-07
M3 Remark V.5 bound (45)	pass	min slack 0.009
M3 q_src correction is necessary	pass	144 cases violate the uncorrected r(e) (min -0.107)
M4 Prop. V.2 thinning (33)	pass	POVM dev 5.6e-17; click-prob spread 0.170 $\rightarrow$ 0.0016 (eta0=0.568)
M4 (35)-(36) dark estimators	pass	unbiased within 4 s.d.; pdc=0: e_sig 0.020 vs 0.0200, pdc=0.001: e_sig 0.020 vs 0.0200, pdc=0.01: e_sig 0.030 vs 0.0300, pdc=0.05: e_sig 0.050 vs 0.0501
M4 (40) cross-talk relation	pass	exact dev 1.2e-16, MC dev 1.1e-03
M4 LP (39) exact = inversion; box gives upward margin	pass	max |LP-inv| 3.1e-16
M5 (47) global tangent; beta-Lambda$\leq$0; beta$\leq$3Lambda/4	pass	min r-tangent -1.1e-16, max(beta-Lambda) -0.019, max(beta-3Lambda/4) -3.8e-04
M5 Cor. VI.3 r_q tangent	pass	min -1.1e-16
M5 App. B range = mu Lambda; (52) variance bound	pass	max Var / bound = 0.900, 0.900, 0.980, 0.900, 0.955
M5 (55)/(74) worst accepted value	pass	max dev 2.2e-16
M6 Table I reproduced	pass	18 entries
M6 Table II reproduced	pass	16 entries
M6 Prop. VII.1 symmetric optimum	pass	free 2-D optimum (pA, pB) = (0.1985, 0.1985), (0.1985, 0.1985), (0.1985, 0.1985)
M6 Prop. VIII.1 exponents (88)-(89)	pass	2*: p -0.335/-0.333, alpha-1 -0.662/-0.667, deficit -0.337/-0.333; 2**: p -0.253/-0.250, alpha-1 -0.761/-0.750, deficit -0.245/-0.250
M7 (69)/(75)-(76) valid upper bounds	pass	MC abort $\leq$ Hoeffding and $\leq$ Bernstein at all L

Check	Result	Evidence
M10 exact abort agrees with MC	pass	max |MC-exact|/sigma 1.52
M10 Hoeffding (69) and Bernstein $\geq$ exact	pass	35 points with exact value above 1e-300
M8 original schedule over-certifies	pass	e=0 while kappa=p, 8 values of p
M8 Protocol 2* detects the same attack	pass	max |e - 3p/4| 0.0021
M8 early J_A counterexample	pass	e = 0, kappa = 1
M8 sifting rates (11), (12), Rem. II.1	pass	max dev 3.9e-04
M9 leftover hash (65), classical toy	pass	48 (leakage, l) pairs, max d/bound 0.984
M11 adversarial search: no violation of (27)	pass	38 optimised attacks, min gap 1.8e-08
M11 optimum reaches r(e) for dim E' $\geq$ 4	pass	max optimised gap 1.4e-05 (dE=4, 8); dE=2 stays $\geq$ 0.025
M11 control: same search breaks the ZZ test	pass	optimised gaps -1.000 to -0.662